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(1) Research Program/Project/Study Title: Effects of Corn and Mungbean Sprouts as Alternative Nutrient Sources on the Growth Performance of Darag Native Chicken (<i>Gallus gallus Domesticus</i>)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)
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		☐ Collaborative Research (external agency)
(7) Sector/Commodity Animal		(8) Discipline/Field of Specialization Animal Nutrition
(9A) Abstract		
(9B) Keywords: Corn, Mungbean, Sprouts, Nutrients, Chemical Composition, Minerals		
(10) Introduction The consumption of chicken meat has experienced exponential growth over the past several decades, making it one of the most widely consumed meats globally. Its popularity is not solely due to its affordability but also to its ability to adapt to diverse culinary traditions and dietary preferences. Its importance as accessible sources of protein, essential vitamins, and minerals for people living in poverty cannot be overstated (FAO, 2013). As global per capita consumption of eggs and white meat continues to rise, the demand for poultry production has surged (Alinaitwe et al., 2018). Chicken farming has emerged as the fastest-growing sector in the livestock industry, and this trend is particularly pronounced in developing countries like the Philippines. However, the soaring demand for poultry production comes with its own set of challenges, most notably the cost of poultry feed. Under intensive production systems, feed expenses can account for a substantial portion of the total production cost, often ranging from 60% to 70%. This financial burden has prompted innovative approaches, such as sprouted fodder production, which is increasingly being adopted in more developed countries (Alinaitwe et al., 2018). Sprouted fodder, in addition to serving as a feed source for livestock during times of fodder scarcity, is also gaining recognition as a suitable poultry feed. Rama et al. (2018) and Chavan and Kadam (1989), noted that seed sprouting undergoes a series of biochemical transformations. These changes encompass the conversion of seed proteins into amino acids, the conversion of carbohydrates into soluble sugars, and the transformation of lipids into essential fatty acids. Furthermore, sprouting triggers an increase in enzyme levels within the seeds, leading to elevated levels of crude protein (CP), ether extract (EE), vitamins, minerals, and improved digestibility. These modifications in nutritional content can have a profound impact on the suitability of sprouted seeds as a nutrient source especially in native chickens that are naturally inclined to thrive on fresh vegetation as their source of nutrients. Energy and protein sources are two fundamental components of animal nutrition, each playing a critical role in the overall health, growth, and productivity of animals. One of the most abundant energy source feedstuff in the Philippines is corn. It is commonly use as primary energy source in animal feed formulations. In a dry matter basis, corn has 3350 kcal/kg metabolizable energy and about 7.5-8% CP. While, the commonly used protein feedstuff is soybean. It is preferred in feed formulations due to its excellent protein content and amino acid profile. However, the production and processing of soybeans can be expensive, contributing to the high cost of soybean-based feeds. This cost can be a significant financial burden for livestock producers, particularly in regions where resources are limited. Recognizing the cost and availability challenges associated with soybean, many feed producers and researchers are exploring alternative protein sources that are more cost-effective and readily available. These alternatives include other commonly available plant protein sources like legumes. The most common legume in the country that can be sprouted is mungbean. The amino acid profile of mungbean sprouts is noteworthy. They contain a range of essential amino acids, which are crucial for animal health and growth. The amino acids in mungbean sprouts can complement those from other feed sources. Despite these claims, there remains a noticeable gap in the available findings, particularly within specific local contexts. It is essential to investigate the ideal germination days of corn that would result in improved		

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growth performance of native chickens specifically, to determine the variation in the nutrient content and percent yield of corn and mungbean sprouts harvested and feed in 3, 6, and 9 days, and its effect in the total feed intake, weight gain, and feed conversion ratio of Darag native chicken; hence, this study is conducted.

(11) Materials and Methods or Methodology

Germination of seeds

The preparation and germination process for the yellow corn seeds and mungbean are adopted from Morgan et al. (1992) wherein, local yellow corn seeds and mungbean, were subjected to an initial cleaning process, by washing it with water to remove impurities and any surface contaminants. Then, the seeds were soaked in a 0.1% hypochlorite solution for 24 hours to disinfects the seeds, reducing the risk of fungal contamination, and promotes the metabolic processes necessary for germination. After the hypochlorite treatment, the seeds were repeatedly washed thoroughly until all impurities and any remaining traces of the hypochlorite solution were removed. Then, the treated seeds were evenly spread out in a 3-layer sprouting deck with a height of 12 inches per deck measuring 118 inches in length by 18 inches width by 3 inches in depth covered with sacks at a seeding rate of 10 kilograms. The sprouts were watered three times daily at 7 am, 12nn, and 4 pm to ensure that the seeds have enough moisture for germination. The sprouting deck was placed under the vacant animal shed covered with mosquito net to prevent the seeds from insect and pests attack. Samplings were done to determine the chemical composition of the unsprouted seeds, 3 days (3d), 6 days (6d), and 9 days (9d) corn and mungbean sprouts through proximate analysis. About 100g of oven-dried samples per treatment were grounded and subjected to the analysis of percent dry matter (%DM), ash (%Ash), CP, EE, crude fiber (%CF), gross energy (GE cal/kg) and percent Calcium (%Ca), phosphorus (%P), magnesium (%Mg), and potassium (%K) were analyzed using Association of Official Analytical Chemists (AOAC) Official Method of Analysis 2016. While, percent yield was determined by dividing the final weight of the corn sprout biomass over the initial weight of seeds multiplied by 100.

Feeding trial

A total of 160 heads six-week old chicks of the Darag native chicken strain were randomly assigned to four dietary treatments replicated four times in a Completely Randomized Design (CRD). Each replicates is composed of 10 chicks that were brood artificially, supplemented with vitamins and mineral and slowly introduced to the experimental diet in their individual cages for 2 weeks as part of the acclimatization period.

Dietary Treatment

The 3d, 6d, and 9d sprouts were air dried using a nylon net bags hanged in a cool, shaded area to remove extra moisture before it was chopped manually to approximately 3-5 mm prior to its incorporation in the feed as treatment diets. The daily dietary requirement of the chicks were divided into 2 parts, 50% sprouts and 50% commercial feeds. The 50% sprouts were proportioned to the different inclusion levels which follows 70% corn sprouts (CS) and 30% mungbean sprouts (MS) for Treatment 2, 60%CS and 40%MS for Treatment 3, and 50%CS and 50%MS for Treatment 4, while, purely commercial feeds was given to the control group assigned as Treatment 1.

Data were analyzed using one-way analysis of variance (ANOVA) in CRD tested for homogeneity using Bartlett's test and normality using Shapiro's test. Trend comparison was done using Tukeys's Honest Significant Difference Test. Results are statistically significant at the 0.05 alpha level.

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(12) Results and Discussions

Chemical variability of nutrients in corn and mungbean before and after several days of sprouting

Table 1 presents the variability in the chemical composition of unsprouted and sprouted corn for 3, 6, and 9 days (d). Percent DM of corn on an as-fed basis showed a highly significant difference ($p < 0.00$), with 9d having the highest DM (98.78%) with no notable deviation from 3d and 6d (97.90% and 98.78%). Similar pattern of differences ($p < 0.01$) was observed in %CP, with the highest increase at 9d (94.46%). While, EE showed a significant difference ($p < 0.00$) with 3d having the highest value (4.27%) followed by 9d and unsprouted with values ranging from 3.60% and 3.62%, respectively. A consistent pattern of variances was observed in the ash content of the samples with 3d having the highest value at 1.59% and 6d having the lowest value at 1.28%. Meanwhile, CF showed a significant increase ($p < 0.00$) from 2.23% in unsprouted corn to 3.28% after 9d of sprouting. Likewise, GE showed a significant increase ($p < 0.00$) with 9d having the highest value at 4,078 cal/kg from 3,922.60 of the unsprouted corn seeds.

As the seed germinates and the seedling starts to grow, it adds new plant tissues such as roots, shoots, and leaves. These new tissues contribute to the increase in dry matter content.

Table 1. Chemical composition of unsprouted and sprouted corn seeds at 3, 6, and 9 days.

Item ¹	Unsprouted seeds	Sprouting day			SEM	P-VALUE
		3d	6d	9d		
DM, %	94.91 ^B	97.76 ^A	97.90 ^A	98.78 ^A	0.39	0.00
CP, %	8.29 ^B	9.08 ^{AB}	8.93 ^{AB}	9.46 ^A	0.15	0.01
EE, %	3.62 ^B	4.27 ^A	3.06 ^C	3.60 ^B	0.11	0.00
Ash, %	1.36 ^B	1.59 ^A	1.28 ^C	1.42 ^B	0.03	0.00
CF, %	2.23 ^C	2.53 ^B	2.77 ^B	3.28 ^A	0.10	0.00
GE, kcal/kg	3,922.60 ^B	4,069.10 ^A	4,003.00 ^A	4,078.00 ^A	0.03	0.00

¹Chemical composition expressed as dry matter (%DM), crude protein (%CP), ether extract (%EE), %ash, and crude fiber (%CF), and gross energy (GE).

-Day (d)

-Means with the same letter superscript in a row are not significantly different

- $P < 0.05$ is significant

Table 2 displays the variances in the chemical composition of mungbean from seeds to 9d sprouting. Percent DM shows a significant difference ($p < 0.00$) with 9d mungbean sprout having the highest DM at 95.58%. A similar pattern of differences was observed in the %CP with a significant increase ($p < 0.00$) from 23.38% in unsprouted mungbean to 36.16% in 9d sprouts. Another significant increase ($p < 0.00$) was detected in the %EE of the mungbean from 0.92% in unsprouted with no remarkable deviation from 3d and 6d having 1.32% and 1.29%, respectively to the highest value of 1.87% in 9d. Meanwhile, a significant increase ($p < 0.00$) was likewise noted in the ash content of the mungbean from 3.65% in unsprouted seeds, 4.53% in 3d, 5.40% in 6d, and the highest value at 6.31% in 9d. A similar pattern of differences was observed in the %CF of mungbean increasing significantly ($p < 0.00$) from 4.09% in unsprouted seeds to 13.65% in 9d. Lastly, GE, the values were significantly decreasing ($p < 0.00$) from 3,721.60 cal/kg of the unsprouted seeds succeeded by 3,632.30 cal/kg for 3d, 3,597 cal/kg for 6d, and 3,507.40 cal/kg for 9d.

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Table 2. Chemical composition of mungbean

¹ Item	Unsprouted seeds	Sprouting day			SEM	P-VALUE
		3d	6d	9d		
DM, %	91.72 ^D	92.69 ^C	94.37 ^B	95.58 ^A	0.39	0.00
CP, %	23.38 ^D	30.39 ^C	33.27 ^B	36.16 ^A	1.23	0.00
EE, %	0.92 ^B	1.32 ^B	1.29 ^B	1.87 ^A	0.89	0.00
Ash, %	3.65 ^D	4.53 ^C	5.40 ^B	6.31 ^A	0.26	0.00
CF, %	4.09 ^D	7.46 ^C	8.46 ^B	13.65 ^A	0.89	0.00
GE, kcal/kg	3,721.60 ^A	3,632.30 ^B	3,597.00 ^C	3,507.40 ^D	0.03	0.00

¹Chemical composition expressed as dry matter (%DM), crude protein (%CP), ether extract (%EE), %ash, and crude fiber (%CF), and gross energy (GE).

-Day (d)

-Means with the same letter superscript in a row are not significantly different

-P<0.05 is significant

Differences in the mineral composition of unsprouted and sprouted corn and mungbean seeds presented in Table 3 showed highly significant differences ($p < 0.00$) across elements measured. Percent Ca found higher in unsprouted corn at 0.33% with measurable similarity in 9d corn sprout, 6d and 9d mungbean sprouts at 0.34%, 0.32%, and 0.42%, respectively. While, unsprouted mungbean have the lowest %Ca at 0.11% with no notable differences from 3d, 6d corn sprouts and 3d sprouted mungbean. On the other hand, %P, %Mg and %K showed highest in 9d mungbean sprouts at 0.63%, 2.36%, and 17.66%, respectively.

Table 3. Mineral content of corn and mungbean sprouts

¹ Item	Corn				Mungbean				SEM	P-Value
	Unsprouted	3d	6d	9d	Unsprouted	3d	6d	9d		
Ca, %	0.33 ^A	0.29 ^{AB}	0.28 ^{AB}	0.34 ^A	0.11 ^B	0.29 ^{AB}	0.32 ^A	0.42 ^A	0.02	0.00
P, %	0.24 ^{DE}	0.25 ^{DE}	0.19 ^F	0.22 ^{EF}	0.28 ^D	0.48 ^C	0.54 ^B	0.63 ^A	0.03	0.00
Mg, %	0.56 ^E	0.54 ^{EF}	0.43 ^{FG}	0.38 ^G	1.29 ^D	1.67 ^C	1.95 ^B	2.36 ^A	0.13	0.00
K, %	4.15 ^E	4.14 ^E	3.20 ^E	3.63 ^E	11.66 ^D	13.67 ^C	16.24 ^B	17.66 ^A	1.04	0.00

¹Mineral composition expressed as Calcium (%Ca), Phosphorus (%P), Magnesium (5Mg) and Potassium (%K)

-Day (d)

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-P<0.05 is significant

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Differences in the growth performance of Darag native chicken fed with corn and mungbean sprouts enriched feed.

Average Feed Intake

Growth Performance of Darag Native Chicken

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Yield and Cost Analysis of Corn and Mungbean Sprouts

Herbage Yield

Growth Performance

